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VODKA-Score: evaluating students' philosophical understanding in allegorical reflections via LLM-distilled QA models

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ABSTRACT

Post-class reflections are a valuable tool for promoting deeper learning and metacognitive development. However, their open-ended and interpretive nature, particularly in subjects like philosophy, makes timely and consistent assessment difficult. This lack of immediate feedback may contribute to learning anxiety and reduce the effectiveness of reflection as a formative practice. In this study, we assume that when a student's reflection is sufficiently complete, it should provide enough semantic information to answer related quiz questions. Based on this idea, we introduce the VODKA-Score, a question-answering (QA) based metric that serves as a proxy indicator, evaluating student understanding by identifying answers within their reflections in response to course-related queries, along with DistilledPlatoBERT, a fine-tuned semantic model distilled from five large language models (LLMs) to generate accurate VODKA-Scores. Experiments with 107 students in a philosophy course show that the best-performing model achieved an accuracy between 0.87 and 0.99 and aligned closely with human scoring under specific conditions, reaching a Cohen's Kappa (κ) above 0.6 and a Spearman correlation (ρ) above 0.5, both statistically significant. Findings suggest that QA-based scoring models can support efficient evaluation in students writing. This study was conducted in a Chinese-language setting and under a fixed reflective writing task, including a predefined prompt, word-count range, writing time, and True/False-based post-test. These conditions limit the cross-linguistic generalizability of the VODKA-Score and should be examined in further studies.

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1. Introduction

Within the reflective practices in supporting student learning development, post-class reflections offer significant educational benefits, as they enable students to critically understand and refine their knowledge in response to feedback, which helps them transcend their previous understanding and develop a deeper comprehension of the course material (X. Zhang, 2023). Reflective writing also enhances metacognitive awareness and self-regulated learning, allowing students to monitor and adjust their learning strategies (Allan & Driscoll, 2014). However, evaluating reflection quality at scale presents substantial challenges: the lack of standardization in assessment methods creates difficulties in establishing reliable and valid evaluation processes across disciplines (Rogers et al., 2019); students often perceive reflection as an ambiguous task, resulting in inconsistent quality and reflections that tend to be non-iterative and monologic rather than the more beneficial iterative and dialogic forms (Tan, 2021); furthermore, studies indicate students frequently submit incomplete reflections, bypassing critical elements such as rationales for future improvements or responses to peer feedback (Katser et al., 2026); Finally, the complexity of the rich but latent meaning of philosophical allegory, which can be observed through an iterative process of knowledge generation in learning activities. Most interpretations and concepts are derived through continuous expression, collision, and linkage between experiences and thoughts, suggesting that the quality of reflective writing is dynamic and evolving, making it more difficult to be captured through static quantification. While instructors are responsible for

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evaluating students' reflections based on contextual considerations and their familiarity with lecture content, standardized assessment tools can support this process.

Automated Assessment Tools (AATs), also known as Automated Grading Systems, have transformed educational evaluation through applications such as automated scoring systems that provide real-time feedback while reducing educator workload (Gardner et al., 2021; Khlaif et al., 2024; Messer et al., 2024). Previous studies have demonstrated the advantages of AATs in the classroom. For instance, Insa et al. (2021) proposed a tool named ASys, which was able to reduce grading time per exam from 6 to 2.5 min and the total grading time from 53.5 to 22.3 h across 535 exams. Mattfeld (2023) applied mastery-based grading in a computer science class, allowing students to submit their assignments repeatedly. Survey results showed that 83% of students reported improved learning motivation, and 94% experienced reduced anxiety. These implementations leverage Natural Language Processing (NLP) and Machine Learning (ML) models to evaluate various forms of student work, including written answers, adaptive tests, and multi-modal assessments (Ouyang et al., 2023). However, significant challenges remain when applying these technologies to reflective writing, which contains nuanced and subjective content that current AI models, including BERT and RoBERTa, struggle to evaluate accurately (Ullmann, 2019; C. Zhang et al., 2024). For specialized domains, researchers have developed subject-specific variants, such as SciBERT, to improve performance by training on relevant datasets (Z. Liu et al., 2023), and yet domains involving humanities or social science concepts present particular difficulties due to the interpretive epistemologies and context-dependent forms of knowledge expression. These limitations underscore the need for innovative approaches to enhance AI capabilities in evaluating complex reflective writing. Meanwhile, inspired by BERTScore (T. Zhang et al., 2019), a scoring system that measures the semantic similarity between documents by applying contextualized embeddings from the pre-trained language models.

We name the generated score the VODKA-Score, as the process of deriving it resembles the distillation process used in making vodka. In this study, we are using Knowledge Distillation (KD) to enable the BERT model to comprehend philosophy in this work. KD is a model compression technique that transfers knowledge from a large, complex model (the teacher) to a smaller, simpler model (the student). This approach is particularly valuable when labeled data is limited, as it allows smaller models to benefit from the rich representations learned by larger models (Fu et al., 2021). Through techniques such as noise purification and pseudo-labelling, KD enables improved performance even with limited labeled samples (Li et al., 2020). In our study, we recognized that KD can be leveraged to transfer specialized philosophical knowledge from large language models (LLMs), such as ChatGPT, to smaller, more efficient Question-Answering (QA) models. Recent advances in contextualized language models such as BERT have made it feasible to assess student writing beyond surface-level features. Its ability to evaluate meaning in context, even in domains like philosophy, supports our semantic understanding-based grading tool. The challenge lies not just in enabling models to understand the literal content of a philosophical allegory, which is the theme of our learning activities design, but in equipping them with the capacity to assess students' interpretive and reflective depth. Moreover, we do not treat VODKA-Score as a direct evaluation of reflective quality. Instead, we conceptualize reflective writing as a knowledge source, and operationalize assessment through whether the reflection contains sufficient semantic information to support correct answers to course-relevant questions, serving as a proxy indicator of conceptual completeness. While KD offer a promising solution to data scarcity in certain domains, the mission is to design an effective workflow that captures deeper philosophical insights within students' reflections. The strategy will be detailed in later sections.

In this work, we propose a metric named VODKA-Score to evaluate students' post-class reflection using QA capabilities as a proxy for learning assessment. The metric is obtained through two phases: the Knowledge Distillation (KD) Phase and the Automated Scoring Phase. Specifically, we utilize student reflections as knowledge sources, expecting them to provide sufficient information to answer course-relevant questions, mirroring how instructors assess understanding. Our proposed approach is based on semantic understanding and philosophical reasoning. It not only examines whether students recall the lecture concepts but also emphasizes whether they can reconstruct the core concept of allegory and build associated links. This kind of tool design will support a more dialogic reflecting process. The approach aims to support teachers by handling large volumes of writing while maintaining precision in evaluating philosophical understanding. To evaluate the effectiveness of our approach within a Chinese-language context and a fixed reflective

writing design with predefined prompts, word-count expectations, time constraints and True/False-based post-test, we define the following research questions:

- **RQ1:** Can a QA-based semantic model, trained via knowledge distillation (KD) from LLMs, reliably evaluate students' post-class philosophical reflections?
- **RQ2:** Under what conditions does the model-generated VODKA-Score align with human-assigned reflection scores in terms of agreement and correlation?

2. Literature review

This study involves two types of models: LLMs for KD, and semantic understanding models for QA. Therefore, in this section, we aim to review relevant literature to inform the design of our experiment and the evaluation methods to be used. We begin with the use of LLMs. A well-crafted prompt is essential for obtaining accurate outputs from LLMs. One widely used strategy is Chain-of-Thought (CoT) prompting, which has proven effective in extracting philosophical knowledge by breaking down complex reasoning tasks into intermediate steps, thereby improving response accuracy (Wei et al., 2022). On the other hand, role-play prompting can enhance interactions with LLMs by guiding the model to adopt specific roles or contextual perspectives, resulting in more accurate outputs. For instance, role-play prompting has been shown to outperform standard zero-shot prompting across multiple datasets, including AQuA and the Last Letter task (Das et al., 2024). Additionally, prompting methods have achieved notable improvements in F1-scores, demonstrating increased precision and recall in transforming raw LLM outputs into structured, query-able knowledge (Kong et al., 2024). When considering verification methods, using multiple LLMs can improve factual validity by reducing hallucinations and fallacious answers (Du et al., 2023). For this study, prompt design will incorporate CoT and role-based strategies to extract philosophical concepts more reliably. Additionally, the extracted knowledge is used to distill conceptual structures that inform a downstream semantic model, allowing it to approximate the quality of students' reflective writing through proxy indicators. The LLM will be guided through carefully constructed prompts that break down Platonic ideas and adopt the role of a philosophy educator to enhance the quality and consistency of extracted knowledge.

Another aspect that deserves attention is the fine-tuning of BERT models, particularly in terms of performance evaluation. A central question is how BERT models can be fine-tuned using LLM-generated data to accurately represent and assess understanding of Platonic concepts. Previous studies, such as BioBERT (Lee et al., 2020) and SciBERT (Beltagy et al., 2019), have shown that successful domain adaptation typically involves initializing with the original BERT model and continuing pre-training on a domain-specific corpus. This approach allows the model to effectively learn specialized language and conceptual structures. In the education domain, Pilicita and Barra (2023) applied several BERT variants to classify educational mobile apps such as math, science, and art. They evaluated performance using Accuracy and F1-score to reflect both correctness and the balance between precision and recall. Similarly, Xu and Zhu (2023) fine-tuned BERT to construct an intelligent question bank for computer science courses, with evaluation metrics including precision, recall, and F1-score. These studies consistently used Accuracy as a core metric because it indicates the proportion of correct predictions, making it a reliable indicator of overall model performance. In educational settings, high accuracy reflects the system's ability to provide appropriate learning content. Therefore, this study adopts Accuracy as the primary metric to evaluate the effectiveness of the proposed BERT model for domain-specific usage.

Prior studies on automated grading have employed diverse systems and evaluation approaches, often utilizing statistical analyses to compare machine-generated scores with those of human graders. For example, Ndukwe et al. (2019) designed a chatbot-based grading system that applied TF-IDF with cosine similarity and the Dandelion API for semantic analysis to score short written answers. They evaluated performance using Cohen's Kappa and reported a value of 0.6, indicating good agreement with human instructors. Similarly, Yang et al. (2021) used BERT-based text summarization to generate marking scores for graduate students, combining these with learning analytics. Their evaluation revealed Kappa values ranging from 0.6 to 0.7 and high correlation coefficients, indicating substantial agreement and a strong trend alignment. In another study, Wu et al. (2023) introduced MeNSP, a zero-shot prompt-based scoring

model for scientific argumentation tasks, which achieved Kappa values ranging from 0.3 to 0.57. With a few-shot approach, the Kappa improved slightly to 0.38, demonstrating moderate agreement even with limited labeled data. Conejo et al. (2018) used Classical Test Theory (CTT) and Item Response Theory (IRT) to assess programming tasks, treating each test case as an item. After removing technical errors, they reported a machine-human Pearson correlation between 0.55 and 0.98, indicating high consistency. Lastly, Liebenberg et al. (2018) evaluated the Fitchfork grading tool using Pearson correlation and found a value of 0.78 ($p < 0.001$), indicating a strong alignment with manual grading but a tendency to underestimate the performance of low-performing students. Overall, Cohen's Kappa and Pearson correlation are widely used to evaluate agreement and score trends in automated grading, with reported values typically ranging from 0.3 to 0.7 for Kappa and 0.55 to 0.98 for correlation. Based on these precedents, our study will also adopt both metrics to assess the consistency and validity of the VODKA-Score.

The last consideration is how to support students in writing meaningful post-class reflections during the learning process. According to Brookfield (2017), effective reflection happens when students are guided to examine their assumptions through multiple perspectives, including personal experience, peer feedback, theoretical understanding, and the learner's point of view. Teachers can assist this process by asking questions such as *What did you assume before this class?*, *What changed your thinking?*, or *How does this connect to course ideas?*. Similarly, Ryan (2013) recommends a structured model with four levels: reporting, relating, reasoning, and reconstructing. To support each level, teachers may ask prompts like *What happened?*, *Have you seen this before?*, or *Why is it important?*. In terms of word count, Brondani (2010) found that a minimum of 150 words was required in a dental education course, with actual submissions ranging from 179 to over 1,000 words depending on the depth of reflection. Regarding time allocation, Larsen et al. (2016) reported that medical students spent an average of 27 min writing each daily reflection during clinical rotations, with a range of 7–90 min. To summarize, this study employs an activity design informed by these findings. Post-class reflections will include structured prompts, require a minimum of 150–300 words, and allow students approximately 20–30 min to complete their writing. The activity setup provides 3–4 focused reflection questions after class, ensures students have quiet time for writing, and emphasizes critical thinking over simple summary.

3. Methodologies

3.1. Participants

We structured the learning activities in a way that guided the content and depth of students' post-reflections, so whether students' performance improved statistically is considered an indicator of the effectiveness of our learning activity. This study involved 107 undergraduate students enrolled in a general education philosophy course. The average age of the participants was 19.4 years, with 58 females and 49 males. The course introduced classical philosophical ideas, and one of the core lecture topics was *Plato's Allegory of the Cave*.

We designed the post-class reflection activity with reference mentioned in Section 2 on effective reflection practices, including (Brondani, 2010; Brookfield, 2017; Larsen et al., 2016; Ryan, 2013). Students were asked to respond to two guiding prompts: (1) *Why is Plato's Allegory of the Cave important?* and (2) *What personal insights or inspiration did you gain from it?* After completing the post-test, students were given 30 min to write a reflection of approximately 150–300 words. While the word count was suggested, it was not enforced strictly, as the emphasis was placed on depth of thinking rather than length.

The writing form among students varies; some write in bullet point format, while some reflections are in the conversational style of writing, highly subjective opinions, or personal experiences. Additionally, there are participants giving extremely short or meaningless responses, which we did not exclude, as the core objective of this study was to evaluate whether the DistilledPlatoBERT could accurately answer True/False test questions based solely on the content of such reflections. The diverse content format showcases the differences regarding the extent to which students comprehend the course content and their interpretive attitudes toward the topic. As for the reflection length shown in Figure 1(a), the average reflection length is 158.0 words. While the majority of students write reflections under 100 words, the distribution is somewhat even between 50 and 300 words. However, the number of reflections declines significantly in the 350–450 word range, and very few exceed 500 words.

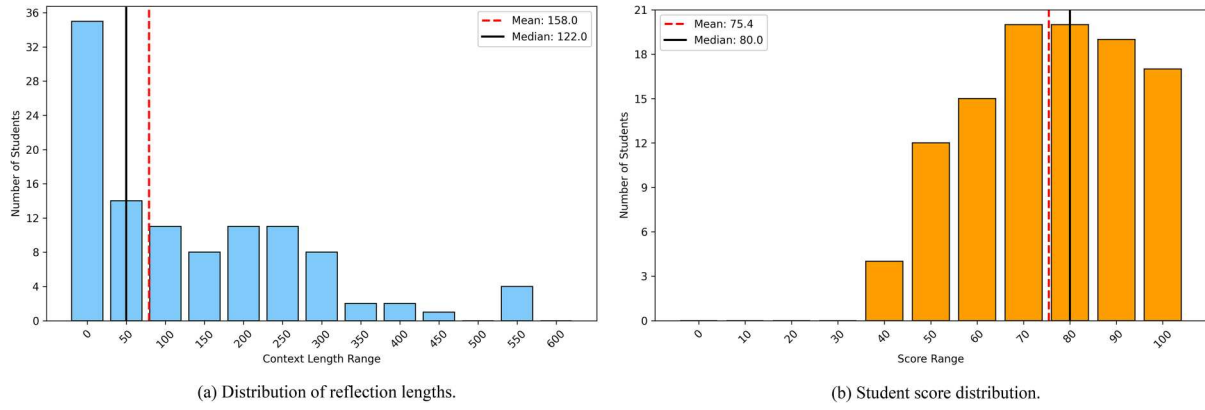


Figure 1. Distributions of the collected dataset from the course. (a) Distribution of reflection lengths. (b) Distribution of student scores.

The last activity is a post-test, which can serve as a formative review of the key lecture content to ensure the content reflects the learning outcomes. We used 10 teacher-designed True/False questions to assess students' learning performance. The distribution in Figure 1(b) shows that most students performed above 60. Specifically, the average score was 75.4, and the median score was 80, indicating that the majority of participants benefited from the learning activities and were able to grasp the key concepts of the lecture.

3.2. Experimental overview

The VODKA-Score proposed in this study is a metric based on the QA Task in NLP. The so-called QA task is a process in which an input question and a context related to the Question are taken, and then the QA model can query the answer in the Context as output. Specifically, if we use the post-class reflection written by students as the Context in the QA task and the post-class quiz designed by the teacher as the Question, then theoretically, if the student's reflection text is complete enough, the content should be strong enough to answer related quiz questions. Otherwise, it will not be able to fully answer all of the questions. Subsequently, a score can be calculated based on the answer correction rate, which is the VODKA-Score proposed in this study. This method is divided into two phases: Knowledge Distillation (KD) phase and Automated Scoring phase. The former is used to prepare the semantic understanding model, while the latter is used to score based on that model. The complete architecture is shown in Figure 2. The following is an explanation of the two phases:

- **Knowledge Distillation (KD) Phase:** A semantic understanding model is required for a specific domain. While Google BERT can be adopted, it is not tailored to the course topic. Therefore, we generate a dataset using an LLM and use it to fine-tune and evaluate the performance of multiple pre-trained BERT models. The dataset generated in this phase is named the Distilled QAC trio, reflecting the QA task requirements of the BERT model: Question, Answer, and Context. The detailed process is described in Section 3.3. The BERT model fine-tuned on the Distilled QAC trio is referred to as DistilledPlatoBERT, with details provided in Section 3.4.
- **Automated Scoring Phase:** We input students' post-class reflections and the corresponding post-class quiz questions into DistilledPlatoBERT to predict their answers. The prediction results are then used to calculate the VODKA-Score. The detailed scoring process is described in Section 3.5. Two evaluations are conducted: one to verify that DistilledPlatoBERT has achieved sufficient semantic understanding of course topics, and another to examine the difference between the VODKA-Score and students' actual scores. These performance evaluation procedures are outlined in Section 3.6.

3.3. Distilled QAC trio

In real-world educational scenarios, courses are typically designed around a professional subject and explore the meaning and purpose behind these specific goals in depth. However, when designing semantic

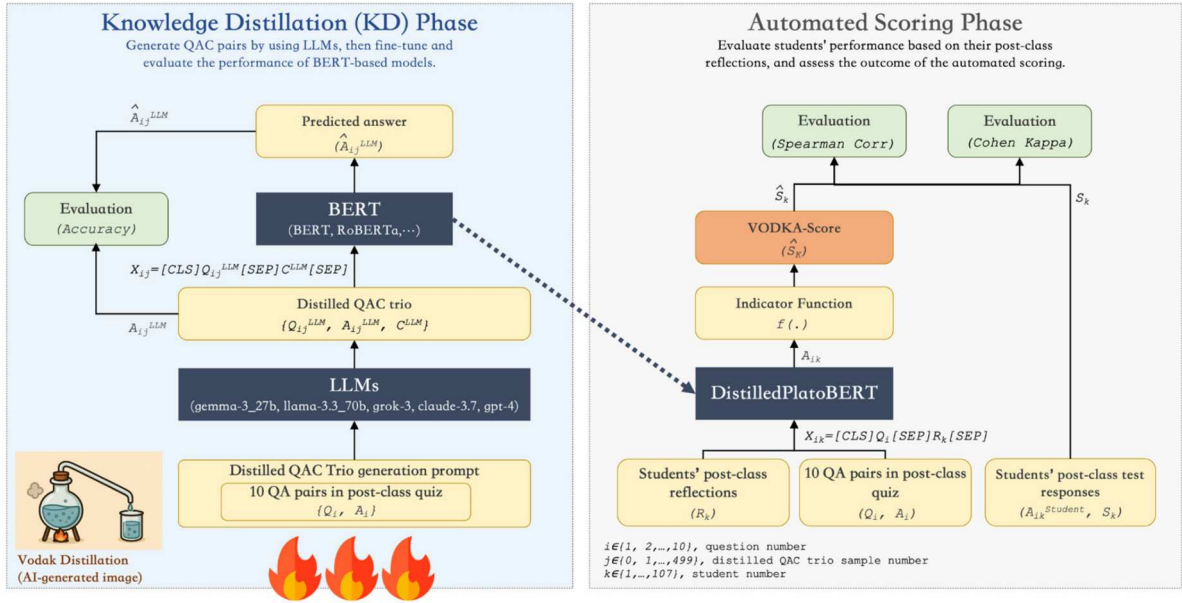


Figure 2. Overview of the proposed VODKA-Score method.

understanding models, the background knowledge of these models is usually insufficient. For example, Google BERT is trained using general data such as Wikipedia and BookCorpus (Devlin et al., 2019). Therefore, this study gathers a dataset in Knowledge Distillation (KD) Phase to enable Google BERT to learn domain-related knowledge from multiple LLMs to become a semantic understanding model dedicated to professional subjects.

More specifically, we gave the LLM 10 true-or-false questions designed by the teacher (for learning outcome evaluation), denoted as, A_i , where i represents the index of the post-class quiz ranging from 0 to 9. The question design mirrors the structure of the learning activities, both of which follow a progression from a surface-level understanding to deeper philosophical reasoning. This progression can be distinguished into three levels. Firstly, a surface-level understanding, concentrating on the background knowledge of the allegory, such as the author-character relationship (Plato and Socrates). At this point, the questions have not yet entered the philosophical reasoning. Subsequently, the representative level focuses on the explicit plot and metaphor in the allegory, for example, inquiring about the meanings of the prisoner, the fire, and the outside world within the allegory. At this level, students can identify and compare the fictional elements to real-world structures. Lastly is the philosophical reasoning level. The questions are designed to guide students in reflecting on how the allegory of the cave can be linked to the educational system, various forms of knowledge, or the perspective of truth. This is the level where critical thinking starts and also with the most space for free interpretation. All the questions are listed in $\{10_QA_pairs\}$ of Table 1. We then prompted the LLM to generate a new dataset based on the concepts in those questions. For each teacher-designed question Q_i the LLM generated 500 semantically related QA pairs, half of which (250) were True and the other half (250) were False, a total of 5,000 samples. Additionally, each sample was accompanied by a concept C^{LLM} , which was written with reference to all ten original questions $Q_1, Q_2, \dots, Q_{10}$ and constrained to within 200 words, aiming to serve as a context capable of answering them all. These generated trios are denoted as $Q_{ij}^{LLM}, A_{ij}^{LLM}, C^{LLM}$, where j is the index of the generated samples for each i ranging from 0 to 499, and the superscript LLM indicates that the data were generated by which specific LLM. The final constructed dataset is referred to as the Distilled QAC trio, defined as:

$$\text{Distilled QAC trio} = \{Q_{ij}^{LLM}, A_{ij}^{LLM}, C^{LLM}\}, \quad i \in \{1, 2, \dots, 10\}, j \in \{0, 1, \dots, 499\} \quad (1)$$

To prompt the LLM to generate the Distilled QAC trio, we refer to Das et al. (2024), Kong et al. (2024), Wei et al. (2022) to design an instruction template as shown in Table 1, which includes five sections: a Role-play section to provide background, 10 teacher-designed post-class QA pairs to define key concepts, a Chain-of-

Table 1. Distilled QAC trio generation prompt.

Strategy	Prompt
Role-play	You are a professional assistant for educational content development. You are skilled at generating high-quality test questions and teaching materials. Please follow the instructions to process the input \${10_QA_pairs} and improve it.
+ 10 QA pairs in post-class	\${10_QA_pairs}: Q₁: Plato's real name was Plato, meaning "broad vision", given because of his deep philosophical insight. A₁: False Q₂: In the *Symposium*, Plato uses Socrates to present the Allegory of the Cave and explore political systems. A₂: False Q₃: In the cave allegory, prisoners believe the shadows and sounds they see and hear are real, showing how people live in ignorance. A₃: True Q₄: The prisoner thinks the cave is more real than the outside world after seeing the sunlight. A₄: True Q₅: Most people are naturally afraid of truth and never adapt to it. A₅: False Q₆: A prisoner who returns to the cave to help others may be punished for doing so. A₆: True Q₇: The allegory symbolizes Socrates' death and shows Plato's belief that education leads people to truth. A₇: False Q₈: The enlightened prisoner feels no envy for leaders inside the cave because he knows the outside truth. A₈: True Q₉: Socrates believed the cave represents the body, and escaping it means finding truth and wisdom. A₉: False Q₁₀: Modern scholars say Plato had two ideas of truth: correctness and uncovering; the second leads people from blindness to clarity. A₁₀: True
+ CoT	Please use a step-by-step approach (Chain-of-Thought) to rewrite two new True/False questions based on the original input: 1. Analyze the original question's difficulty, clarity, and educational value. 2. Consider how to make the question clearer, more challenging, or better at testing understanding. 3. Evaluate the original answer for completeness, correctness, and educational relevance. 4. The first new question should be a true statement (answer: 1). 5. The second new question should be a false statement (answer: 0). 6. Based on the reasoning above, rewrite both the questions and their answers. 7. Ensure that the context can respond to the concepts and ideas implied in all the questions: \${10_QA_pairs}.
+ One-shot	Please return only a new QAC trio in JSON format. The format should be: <pre>[{"question": "question text", "answer": 1, "context": "Plato's Allegory of the Cave uses the image of prisoners chained inside a cave, only able to see shadows, to describe how humans come to understand truth...", "question": "question text", "answer": 1, "context": "In the story, the cave represents the world of appearances, the shadows are the false reality people believe in, and the sunlight stands for the light of truth..."}]</pre>
+ Output format	Additional rules: 1. The output must be in Traditional Chinese. 2. The questions must be in True/False format. 3. Do not include any explanations. 4. The Context should be a single paragraph, within 200 words.

Thought (CoT) component to improve output accuracy, a One-shot example, and an Output format specification. Each output is formatted in JSON with three fields: *Question*, *Answer*, and *Context*.

In this phase, we obtained the Distilled QAC trio from several state-of-the-art LLMs, denoted as superscript LLM in equation (1), including two open-source models: Google/Gemme-3 (hereafter: *gemma-3_27b*) (Team et al., 2025) and Meta/LLaMA-3.3 (hereafter: *llama-3.3_70b*) (Grattafiori et al., 2024), as well as three commercial models: xAI/grok-3-latest (hereafter: *grok-3*) (xAI, 2025), Anthropic/claude-3-7-sonnet (hereafter: *claude-3.7*) (Anthropic, 2025), and OpenAI/gpt-4 (hereafter: *gpt-4*) (Achiam et al., 2023). Through the above process, we summarized the statistical characteristics of the data generated by LLM in Section 3.1, and followed up usage in Section 3.4.

3.4. DistilledPlatoBERT

DistilledPlatoBERT aims to assess the presence of truth-relevant reasoning in student-written reflections or concept passages by determining whether they contain adequate support to answer given philosophical questions. It is a philosophy domain-specific, BERT model fine-tuned for binary classification tasks, specifically designed to predict True/False questions associated with philosophical concepts. The model is trained using the Distilled QAC trio dataset generated by LLMs, grounded in classic philosophical ideas such as *Plato's Allegory of the Cave*. This fine-tuning process not only grants the model the ability to understand philosophical concepts, but also customizes the input and output format to accept teacher-designed questions and students' post-class reflections as input and the answer as output.

To predict the True/False answer from reflection text and teacher-designed questions, we formulate the fine-tuning process as a binary classification task, as shown in equations (2) and (3). First, in equation (2), the

input X_{ij} consists of a question Q_{ij}^{LLM} , and a context C^{LLM} , which correspond to the LLM generated questions and LLM generated reflections, respectively. We use the $[CLS]$ token as the representation for classification, and the $[SEP]$ token as the separator between the question and the context. In both equations, i denotes the question number (ranging from 1 to 10) and j denotes the sample number in the Distilled QAC trio dataset (ranging from 0 to 499). The superscript LLM indicates which specific LLM generated the data which listed in Section 3.3.

$$X_{ij} = [CLS]Q_{ij}^{LLM}[SEP]C^{LLM}[SEP], \quad i \in \{1, 2, \dots, 10\}, j \in \{1, 2, \dots, 499\} \quad (2)$$

On the other hand, the goal of this fine-tuned model is to predict whether the $\hat{A}_{ij}^{LLM}$ is the predicted True or False answer corresponding to Q_{ij}^{LLM} and C^{LLM} , where the tuning procedure is presented in equation (3). The model applies a linear layer to the $[CLS]$ representation to produce a two-dimensional output vector z_{ij} . The softmax function is then used to calculate the class probabilities, and the predicted label $\hat{A}_{ij}^{LLM}$ is selected as the class with the highest probability. Finally, $\hat{A}_{ij}^{LLM}$ is expected to form a sequence of True and False values, and we calculate the VODKA-Score based on this sequence of binary results. Further details can be found in Section 3.5.

$$z_{ij} = Wh_{[CLS]} + b, \quad P_{ij} = \text{softmax}(z_{ij}), \quad \hat{A}_{ij}^{LLM} = \text{argmax}(P_{ij}) \quad (3)$$

In this phase, we obtain DistilledPlatoBERT based on a fine-tuning process applied to several pre-trained BERT variants, including `google/bert-base-uncased` (Devlin et al., 2019), `google/roberta-base` (Y. Liu et al., 2019), `google/bert-base-chinese`,¹ `ckiplab/bert-base-chinese`,² `hfl/chinese-macbert-large` (Cui et al., 2020), and `weiweishi/roc-bert-base-zh` (Su et al., 2022). These models can be broadly categorized into two groups: native Google models and models specifically optimized for Chinese language tasks. In addition, because the LLM and BERT models used in this study can be combined to fine-tune a total of 30 philosophy-domain-specific models, we aim to identify a subset of promising combinations to be referred to as DistilledPlatoBERT. In this study, we must ensure that DistilledPlatoBERT has a comprehensive understanding of philosophical semantics; to this end, we will explain the evaluation metrics in Section 3.6.1.

3.5. VODKA-Score: $\hat{S}_k$

In the Automated Scoring Phase, we calculate students' VODKA-Score based on the prediction results of DistilledPlatoBERT. The VODKA-Score is intended to serve as a reference score for evaluating the quality of students' reflections. The underlying concept is to use the model to infer answers to post-class questions designed by teachers, using the reflections written by students as input. Assuming satisfactory model performance, the higher the number of correctly inferred questions, the higher the score; otherwise, the score will be lower.

To calculate the VODKA-Score, details can refer to equations (4) and (5). First, in equation (4), the input to DistilledPlatoBERT is similar to the fine-tuning process described in Section 3.4, which contains two elements: a question Q_i and a reflection text R_k , corresponding to the teacher-designed questions and students' post-class reflections, respectively. The $[CLS]$ and $[SEP]$ tokens are also used as the representation for classification and as the separator. In both equations, i denotes the question number (ranging from 1 to 10), and k denotes the student number (ranging from 1 to 107). $\hat{A}_{ik}$ is the predicted True or False answer corresponding to Q_i and R_k .

$$\hat{A}_{ik} = \text{DistilledPlatoBERT}([CLS]Q_i[SEP]R_k[SEP]), \quad i \in \{1, 2, \dots, 10\}, k \in \{1, 2, \dots, 107\} \quad (4)$$

Since $\hat{A}_{ik}$ is a series of prediction answers with True or False values, we apply an indicator function, denoted as $I(\cdot)$ in equation (5), to convert the prediction results into a number from 0 to 100 with an interval of 10. In practice, we compare $\hat{A}_{ik}$, predicted by the model, with the actual answer A_i provided by the teacher for each question. If $\hat{A}_{ik}$ is equal to A_i , the indicator function returns 1; otherwise, it returns 0. We then calculate the sum of the indicator function's outputs for the 10 questions and multiply it by 10 to obtain $\hat{S}_k$. Ideally, if the student's reflection is complete and the model predicts all 10 answers correctly, then $\hat{S}_k$ will be a full score. Conversely, if the reflection content is incomplete and the model predicts all 10 answers

incorrectly, then $\hat{S}_k$ will be 0. In this equation, $\hat{S}_k$ is the VODKA-Score we proposed in this study, and we will define the evaluation metrics in Section 3.6.2.

$$\hat{S}_k = \sum_{i=1}^{10} I(A_i = \hat{A}_{ik}) \times 10 \quad (5)$$

3.6. Evaluation metrics

The evaluation of this work covers two phases, as shown in Figure 2. First, we fine-tuned DistilledPlatoBERT using the Distilled QAC trio dataset generated by several LLMs, with a particular focus on *Plato's Allegory of the Cave*. We must ensure the accuracy of this model, specifically its ability to understand semantics, in order to support its use in the second phase. Second, we used DistilledPlatoBERT to query students' post-class reflections in order to answer the questions in the post-class test and to calculate the VODKA-Score based on the model's prediction results. This score needs to be evaluated by measuring the level of agreement and correlation with the students' actual scores. Below, we explain the evaluation metrics and their relevance to the proposed scoring process:

3.6.1. Evaluation metric for DistilledPlatoBERT

In the Knowledge Distillation (KD) Phase, we used 80% allocated in Distilled QAC trio for fine-tuning and 20% for validation. Since the indices of the dataset have no meaning, we first randomly shuffled the data before splitting. To ensure robust evaluation, we repeated the random 80/20 train-validation split ten times, each time re-shuffling the dataset and recording performance. The final results were averaged across the ten runs to mitigate sampling bias and enhance the reliability of the evaluation.

To evaluate the performance of DistilledPlatoBERT, we refer to Pilicita and Barra (2023) and Xu and Zhu (2023), which employed BERT models in educational settings. We compute the classification accuracy based on its predicted answers $\hat{A}_{ij}^{LLM}$ and the corresponding ground truth labels A_{ij}^{LLM} . The accuracy is defined as equation (6). Here, i denotes the index of the teacher-designed True/False questions (ranging from 1 to 10), and j represents the sample index in the 20% held-out validation set, with ($N = 1000$). The indicator function $I(\cdot)$ returns 1 if the model's prediction matches the correct label and 0 otherwise.

$$\text{Accuracy} = \frac{1}{N} \sum_{j=1}^N I(\hat{A}_{ij}^{LLM} = A_{ij}^{LLM}) \quad (6)$$

The final accuracy is reported as the average over the 10 runs. Ideally, if the model answers all validation samples correctly, the accuracy would be 100%, indicating a strong understanding of philosophical concepts. Conversely, a poorly performing model would produce an accuracy close to 0%. Later, we will explain the experimental results in Section 4.1.

3.6.2. Evaluation metrics for VODKA-Score

In the Automated Scoring Phase, we refer to Conejo et al. (2018), Liebenberg et al. (2018), Ndukwe et al. (2019), Wu et al. (2023) and Yang et al. (2021) to apply the metric Cohen's Kappa and Correlation for evaluation. However, we obtain the VODKA-Score, which ranges from 0 to 100 in intervals of 10. It is treated as a discrete numerical variable. Therefore, we use the *Quadratic Weighted Cohen's Kappa* (hereafter: κ) and *Spearman Correlation* (hereafter: ρ) to evaluate the agreement and consistency between the actual scores and the VODKA-Scores.

We used the κ , as defined in equations (7) and (8) to calculate the agreement between the VODKA-Score $\hat{S}_k$ and the actual student scores S_k . Both scores are discretized into 11 categories ranging from 0 to 100 in intervals of 10. For each student $k \in \{1, \dots, K\}$, $\hat{S}_k$ and S_k are mapped to their corresponding category indices n and m , respectively, where n denotes the predicted score category and m denotes the actual score category. The observed agreement is represented by the confusion matrix $O_{\{m,n\}}$, and the expected agreement under chance is denoted by $E_{\{m,n\}}$. The weight $w_{\{m,n\}}$ applies a quadratic penalty based on the

distance between categories, where $N = 11$ is the total number of categories.

$$\kappa = 1 - \frac{\sum_{m,n} w_{m,n} O_{m,n}}{\sum_{m,n} w_{m,n} E_{m,n}} \quad (7)$$

$$w_{m,n} = \frac{(m - n)^2}{(N - 1)^2} \quad (8)$$

In addition to agreement, we used ρ , as defined in equation (9), to evaluate the consistency between the VODKA-Score $\hat{S}_k$ and the actual student scores S_k . Both scores range from 0 to 100 and are treated as continuous variables without discretization. For each student $k \in \{1, \dots, K\}$, we compute the ranks $\hat{r}_k$ and r_k corresponding to $\hat{S}_k$ and S_k , respectively.

$$\rho = 1 - \frac{6 \sum_{k=1}^K (\hat{r}_k - r_k)^2}{K(K^2 - 1)} \quad (9)$$

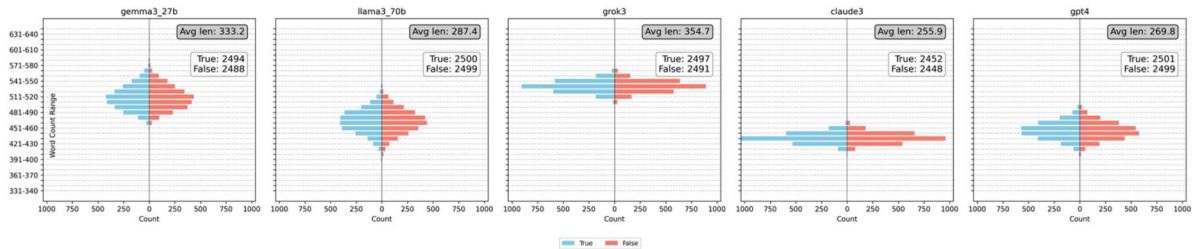
The κ will be divided into several levels, such as 1 for perfect agreement, 0.81–0.99 for near perfect agreement, and 0.61–0.80 for substantial agreement. The ρ will also be divided into several levels, such as 0.9–1 for high correlation, 0.79–0.89 for high correlation, and 0.5–0.69 for moderate correlation, accompanied by a p -value to indicate statistical significance. Later, we will explain the experimental results in Section 4.2.

4. Results

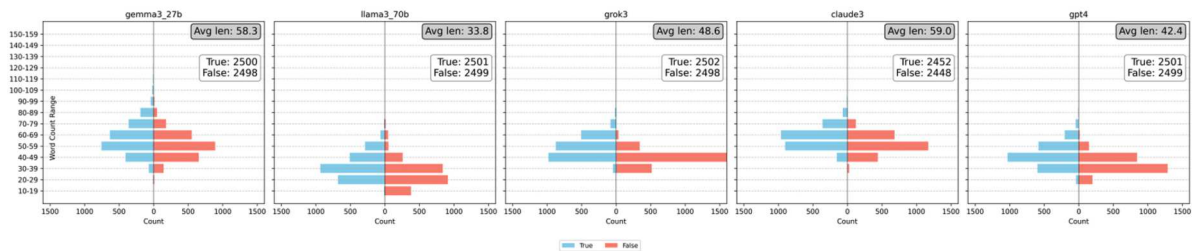
4.1. Evaluation of DistilledPlatoBERT for accuracy

To evaluate the effectiveness of Knowledge Distillation (KD) Phase, which transfers specialized philosophical knowledge from LLMs such as GPT to smaller QA models named DistilledPlatoBERT (**RQ1**: Can a QA-based semantic model, trained via knowledge distillation from LLMs, reliably evaluate students' post-class philosophical reflections?), we summarized the experimental results using the metrics defined in Section 3.6.1.

Descriptive statistics are compiled to summarize the training dataset's characteristics. As displayed in Figure 3, all models have produced 5,000 samples, and each model's output maintains a balanced distribution between True and False labels. The table shows the average length of generated contexts and



(a) Distribution in context outputs



(b) Distribution in question outputs

Figure 3. Comparison of response length and label distributions across models. (a) Distribution in context outputs. (b) Distribution in question outputs.

questions. Among all models, `grok-3` produced the longest contexts on average, with a mean length of 356.6 words, whereas `claude-3.7` produced the shortest, with a mean length of 255.96 words. Regarding the length of questions, `gemma-3_27b` and `claude-3.7`'s outputs are longer, with a mean of 58 words. `llama-3.3_70b` and `gpt-4` generated relatively shorter texts, with an average length of between 30 and 40 words.

Looking into the distribution showcased in each figure, the first one is the question length distribution. Most models' outputs are centered around 40–80 words. `gemma-3_27b` and `claude-3.7`'s question outputs are more spread out, with some being over 100 words. In comparison, `gpt-4` and `llama-3.3_70b`'s output length is short, concentrating on 30–60 words. The second figure displays the context length distribution. Five models' context outputs are located around 400–500 words. The distribution appears more symmetrical, indicating that the language models' generation at this stage is relatively stable and consistent.

The fine-tuning accuracy reported in Table 2 reflects the performance of six BERT-based QA models after being fine-tuned on distilled knowledge generated by five LLMs. It is important to note that high fine-tuning accuracy does not necessarily indicate that a model qualifies as a DistilledPlatoBERT. What defines it is the model's performance to produce accurate VODKA-Score when evaluating students' philosophical reflections.

All pre-trained BERT models have improved significantly after fine-tuning and mostly reached above 0.90 in accuracy. Looking at the fine-tuning performances of the seven BERT models across different LLMs (as displayed on the vertical axis), `grok-3`'s output achieved nearly perfect accuracy of 0.99, followed by `claude-3.7` with an average accuracy of 0.96. By contrast, the fine-tuning data generated by `gemma-3_27b`, `llama-3.3_70b`, and others appears to result in more variable results. In particular, for non-Chinese BERT models such as `google/bert-base-uncased` and `google/roberta-base`, they tend to perform slightly worse across all LLMs. For example, the fine-tuned accuracy of `gemma-3_27b` + `google/bert-base-uncased` is 0.75, while that of `gpt-4` + `google/roberta-base` is 0.86, making them the lowest-performing BERT combinations under their respective LLMs.

We noticed that LLMs with nearly perfect fine-tuning performance (such as `grok-3`) tend to generate output with a narrower word count distribution (as seen in Figure 3). While the accuracy is high, we assumed that the semantic diversity of such training data might be low. To examine this, we computed the Shannon entropy for both the context and question of every LLM's output. As the result shows in Table 3, `grok-3` is ranked second in context entropy with a score of 7.64, and last in question entropy with a score of 7.49.

4.2. Evaluation of VODKA-Score with Kappa (κ) and Correlation (ρ)

To validate the real-world performance of the QA model DistilledPlatoBERT and the VODKA-Score in the Automated Scoring Phase within the context of a philosophy course (**RQ2**: Under what conditions does

Table 2. BERT model's ability to understand the *Allegory of the Cave* (pre-training accuracy/fine-tuning accuracy).

BERT models	LLMs				
	<code>gemma-3_27b</code>	<code>llama-3.3_70b</code>	<code>grok-3</code>	<code>gpt-4</code>	<code>claude-3.7</code>
<code>google/bert-base-uncased</code>	0.50/0.75	0.50/0.90	0.50/0.98	0.50/0.87	0.50/0.93
<code>google/roberta-base</code>	0.52/0.78	0.50/0.91	0.50/0.99	0.50/0.86	0.50/0.95
<code>google/bert-base-chinese</code>	0.49/0.95	0.49/0.97	0.50/0.99	0.50/0.98	0.50/0.98
<code>ckiplab/bert-base-chinese</code>	0.49/0.93	0.50/0.97	0.50/0.99	0.51/0.97	0.50/0.99
<code>hfl/chinese-macbert-large</code>	0.50/0.95	0.51 / 0.97	0.51/0.99	0.52/0.98	0.50/0.99
<code>weiweishi/roc-bert-base-zh</code>	0.49/0.91	0.51/0.94	0.47/0.99	0.50/0.93	0.50/0.91
	0.50/0.87	0.50/0.94	0.50/0.99	0.50/0.93	0.50/0.96

Table 3. Entropy of generated content and question across different LLMs.

	LLMs				
	<code>gemma-3_27b</code>	<code>llama-3.3_70b</code>	<code>grok-3</code>	<code>gpt-4</code>	<code>claude-3.7</code>
Generated context	8.02	7.50	7.64	7.51	7.50
Generated question	8.53	7.69	7.49	7.68	7.77

the model-generated VODKA-Score align with human-assigned reflection scores in terms of agreement and correlation?), we conducted an experiment following the metrics described in Section 3.6.2. The evaluation focused on assessing how well the model-generated scores align with human-graded student performance based on their post-class reflections. The results of this experiment are visualized in Figure 4.

Figure 4 includes five subfigures. The subfigures display the performance of each BERT model under a specific LLM’s fine-tuning dataset and their corresponding κ across different ranges of students’ reflection word counts. The 0–300 on the x -axis presents the word count of students’ reflection. Reflections exceeding 300 words are grouped into the 300 words interval due to their low frequency. The y -axis is the κ , indicating the level of agreement between students’ post-test scores and VODKA-Score. Regarding the annotation text

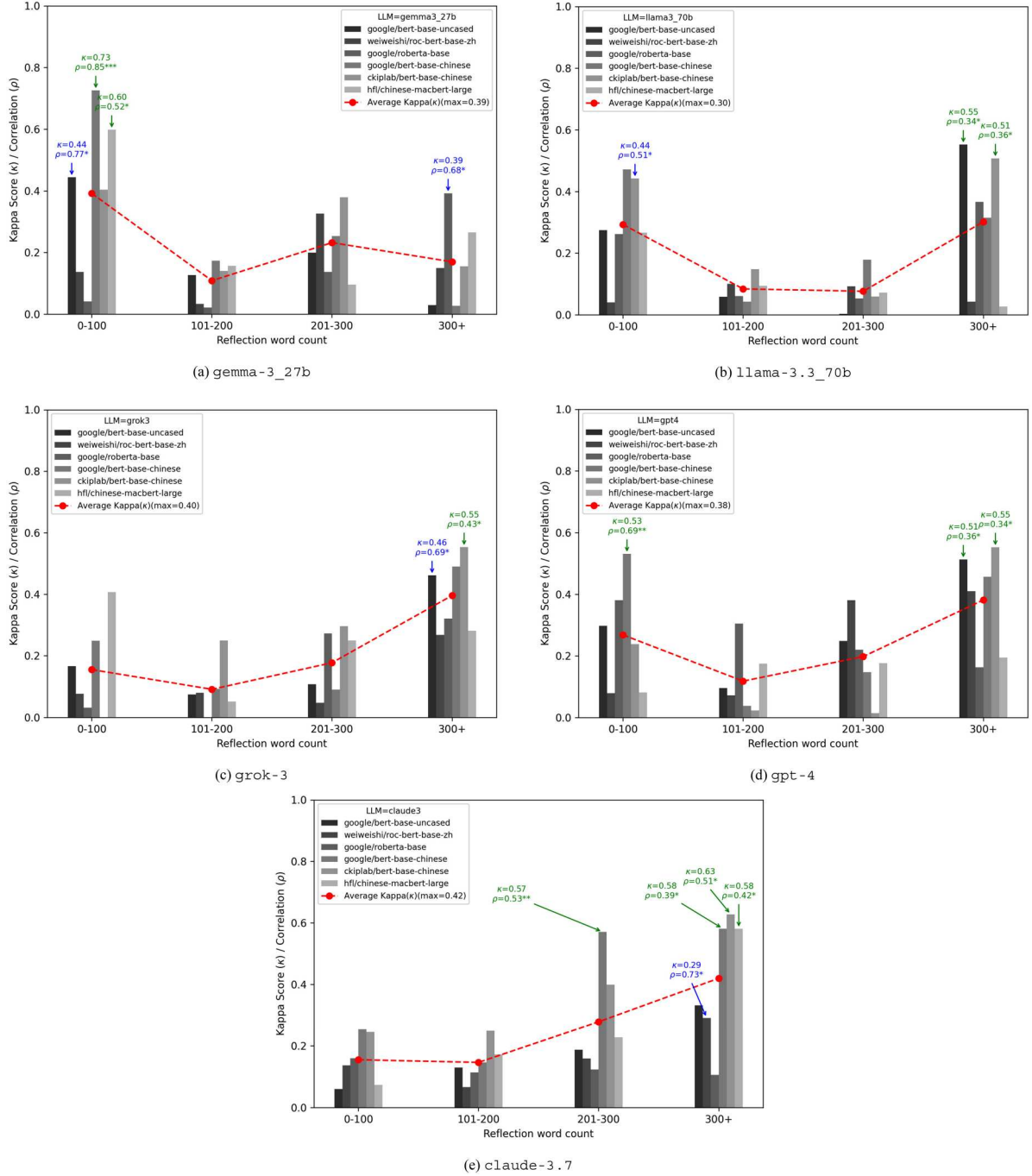


Figure 4. κ and ρ between VODKA-Score and students’ actual scores across different context lengths. (a) gemma-3_27b. (b) llama-3.3_70b. (c) grok-3. (d) gpt-4. (e) claude-3.7

in each subfigure, the red dotted line represents the trend line for the average κ of all models. The colored text is the highlight annotation, with blue text indicating cases where the correlation between VODKA-Score and actual scores exceeds 0.5. In contrast, the green text labels highlight instances where both the correlation is greater than 0.5 and the κ surpasses 0.6, reflecting stronger agreement and consistency in those conditions.

Examining the overall trend in Figure 4, all LLMs perform significantly better within the 300-word count interval, indicating that long text leads to higher agreement between VODKA-Score and actual scores. Performance in the 0–100 range is unstable, and yet `gemma-3_27b` stands out by achieving strong results in this challenging range, especially when paired with `google/bert-base-chinese`. As shown in Figure 4(a), it reaches a κ of 0.73 and ρ of 0.85, indicating substantial agreement and high correlation. The underlying reason may be traced to the entropy analysis in Table 3, where `gemma-3_27b` exhibits particularly high entropy, suggesting that the contexts and questions it generates are richer and more informative even in shorter texts.

In the middle range, consistency is generally low, and the average κ tends to be the lowest across all models. The second-best performance can be seen in Figure 4(e). `claude-3.7`, as the last one in the subfigure, shows the most green labels among all models, particularly in the 300+ range. It performed best in the `ckiplab/bert-base-chinese` combination, where it reached a κ of 0.63, surpassing the 0.6 threshold for substantial agreement, and a moderate ρ of 0.51. While `claude-3.7` has the highest average κ , the best combinations are still observed in `gemma-3_27b`, `llama-3.3_70b` in Figure 4(b), `grok-3` in Figure 4(c) and `gpt-4` in Figure 4(d) all show a U-shaped trend with higher values at both ends and a dip in the 101–300 range. The performance dip observed in this range requires closer inspection and qualitative interpretation, which are discussed in Section 5.

To enable a more nuanced comparison between the VODKA-Score and students' actual performance, we present two score distribution plots in Figure 5, where the x-axis represents the score range and the y-axis represents the number of students. Figure 5(a) shows the VODKA-Score results using the model combination `claude-3.7 + google/bert-base-chinese`. This combination yielded high κ and ρ values when students' reflections exceeded 300 words. In the graph, the black line shows the students' actual scores and the red dotted line shows the VODKA-Score. The two lines follow a similar trend, mostly falling within the score range of 40–80, but exhibit a slight prediction error around the 60 range. There are fewer scores in the 90–100 range, indicating that the predictions tend to underestimate student performance in the higher score range. On the other hand, Figure 5(b) shows the result of `grok-3 + google/roberta-base`, which is one of the combinations that shows greater deviation from the actual scores. In this figure, the black line and red dotted line again show the actual scores and VODKA-Score, respectively. Most of the VODKA-Score (blue bars) are skewed toward the 30–60 ranges, suggesting that the model tends to underestimate students' performance, and only a few students get scores in the 70–80 range, with no grading falling into the 90–100 range, this combination fails to detect any high-performing students.

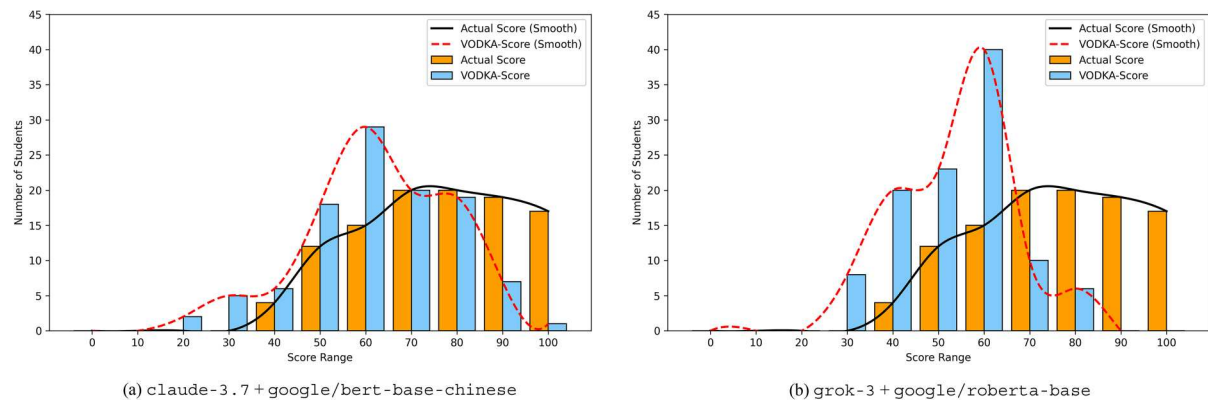


Figure 5. Distributions of VODKA-Score and students' actual scores. (a) `claude-3.7 + google/bert-base-chinese`. (b) `grok-3 + google/roberta-base`

5. Discussions

As mentioned in Section 3.4, our current experiments involve five LLMs and six BERT-based semantic models, resulting in 30 trained models through different pairwise combinations. The LLMs were used to generate training data, while the BERT-based models serve as the scorers in the automated evaluation process. Among the tested model combinations, the VODKA-Score of `claude-3.7+google/bert-base-chinese` most closely match the actual scores. In contrast, `grok-3+google/roberta-base` produces predictions that diverge from the actual scores, with the VODKA-Score's distribution following a trend that is completely reversed from the actual distribution. Although the `gemma-3.27b+google/bert-base-chinese` combination achieved the highest agreement and correlation in the shortest reflection range (0–100 words), our final selection prioritized robustness and stability across reflection lengths rather than peak performance in a single interval. This decision reflects the intended instructional use of post-class reflections, where reflection length naturally varies across students and cannot be constrained to short responses in authentic classroom settings. Moreover, as shown in Table 3, `gemma-3_27b` exhibits substantially higher entropy (8.02–8.53) compared to `claude-3.7` (7.50–7.77). The higher entropy contributes to richer cues in very short reflections, it also introduces greater variability, resulting in less consistent behavior across mid- and long-length reflections. Therefore, we identify `claude-3.7+google/bert-base-chinese` to be DistilledPlatoBERT due to its stable performance on semantic comprehension and grading abilities that achieved the intended purpose of this method.

Previous BERT applications typically rely on human-annotated datasets for training and evaluation, such as BioBERT (Lee et al., 2020) and reflective writing classification (C. Zhang et al., 2024). The promising performance of DistilledPlatoBERT suggests that KD can serve as a feasible alternative to the time-consuming and labor-intensive process. Before further discussing the implications of the DistilledPlatoBERT's applicability, it is important to distinguish the nature of the scoring logic underlying this work. VODKA-Score, a high-purity outcome extracted through multiple stages of refinement, is not used to detect specific keywords or paraphrase the lecture content, but to assess comprehension of the deeper meaning. The aim is to check that students reconstruct concepts throughout and establish a logical and comprehensive structure. We leverage DistilledPlatoBERT to determine if each writing can be used to answer questions related to philosophical essences. KD is used to generate surface-to-deep facets of the allegory in question-and-answer patterns, such as explicit symbolic references and character roles that reveal deeper conceptual elements. Through the scattered but varied distilled knowledge, BERT models learn to reconstruct the whole reasoning behind allegory and can identify and assess deeper meanings in student reflections.

The results indicate that under specific conditions, the VODKA-Score aligns closely with human grading and meets established benchmarks from prior studies. However, upon closer inspection of the results reported a U-shaped performance pattern in Section 4 and examined the reflections within this range. One example is shown below, where a student's reflection received a relatively low human-assigned post-test score (50) but a substantially higher VODKA-Score (90). As demonstrated in the ChatGPT translation, the student's response focuses primarily on the general functions and social roles of education, rather than on the philosophical structure or key elements of the allegory.

I came to understand the real meaning of education and learning in everyday life. On the surface, learning is about studying knowledge from books and academic subjects, but in reality, it is about using education to place people with different personalities into different fields. Because everyone has something they are good at, the role of education is to teach people so that they can know what kind of place suits them and how to make use of it ...

Despite the conceptual misalignment, the reflection contains coherent language that provides sufficient semantic cues for the QA-based model, resulting in a high VODKA-Score despite its limited philosophical depth from a human grading perspective. This case illustrates how mid-length reflections can be semantically answerable but conceptually incomplete, resulting in a dip in performance within the U-shaped trend.

These results can be interpreted as the Knowledge Distillation (KD) Phase is both feasible and effective in Chinese-language applications, echoing the findings of Li et al. (2020) on the usefulness of KD in limited-label scenarios. In the Automated Scoring Phase (Section 4-2), we find that some model combinations perform exceptionally well in specific length intervals, demonstrating their potential as effective instances of DistilledPlatoBERT. These findings indicate that our approach is not only technically viable but also aligned with

the goals of reflective writing. The results that models can reliably assess in philosophical reflections suggest that reflection quality can be captured by QA-based semantic understanding, highlighting whether student responses touch upon key conceptual elements associated with metacognitive awareness and deeper comprehension, as discussed in Allan and Driscoll (2014) and X. Zhang (2023). For students, the VODKA-Score may help alleviate students' anxiety (Mattfeld, 2023) by providing timely, consistent, and interpretable feedback that clarifies what constitutes a high-quality reflection.

The VODKA-Score is a formative support tool within instructional workflows. In practice, the proposed system is designed to support the evaluation of post-class reflections. Hence, the primary use case of the VODKA-Score is situated in the post-class stage. Once reflections are submitted, the system can generate VODKA-Scores, allowing teachers to identify reflections in which core concepts are insufficiently constructed promptly. Rather than requiring them to review all submissions in detail manually, the score functions as an early signal that helps prioritize instructional attention, enabling timely and targeted qualitative feedback. However, to avoid ethical risks associated with over-reliance on this automation technology, the VODKA-Score should inform both instructors and students as a proxy indicator rather than a definitive judgment. Both instructors and students are encouraged to interpret AI-generated scores responsibly and recognize that the model's answerability-based assessment of text sentiment is not the full depth or epistemic value of reflective thinking.

We acknowledge three limitations of the study. First, the collected post-class reflections were produced under clear instructional guidance and teacher-provided hints. As a result, the observed outcomes may depend in part on this guidance. Future work should therefore examine how variations in instructional design influence students' performance and the model's evaluation results. Second, the study is constrained by language and task design. The learning activities were conducted in Chinese and relied on a fixed reflective writing format, including predefined prompts, word-count expectations, and time allocation, as specified in Section 3.1. Third, the experiment relied on ten teacher-designed True/False post-test questions as ground truth. While these results provide an apparent validation reference, the absence of post-test scores would leave the current design without a critical validation component. Future studies should explore more flexible and ecologically valid ground-truth measures in classroom practice to enable continuous tracking of student learning over time. Beyond these methodological limitations, future research should also examine the behavioral effects of automated formative feedback. As noted by Mattfeld (2023), students often experience anxiety due to uncertainty about how their reflective writing will be graded. Providing timely feedback may help alleviate such anxiety. Experimental comparisons could further investigate student performance and self-adjustment with and without automated tool support.

6. Conclusions

We proactively adjust the technical workflow specifically in response to the highly abstract nature of the philosophy discipline or modify the method to deal with manually labeled issues. These approaches are not intended to adapt philosophical education to AI but rather to enable AI to understand the knowledge structure of philosophy, thereby fostering the spirit of intersubjectivity in digital humanities. Importantly, the VODKA-Score is not intended to represent the quality of philosophical reflection but rather to serve as a proxy indicator of whether a student's reflection contains sufficient semantic structure to support course-relevant reasoning. The process also raises thought-provoking observations. Firstly, it is noted that the role of reflection writing changes throughout the workflow progression. As the tool aims to provide real-time feedback that informs subsequent actions such as revising ideas, rewriting, or engaging in discussion, reflective writing becomes a dynamic node in the learning process rather than a fixed text subject to grading, shifting from being an object of evaluation to an integral part of teaching itself and moving its position in the knowledge workflow from the output stage to the formative stage. Subsequently, in the discussion, we mentioned that future work should focus on how the tools are being instructed into actual educational sites, including how teachers use the system, how students respond, whether there are meaningful revisions and reflections based on the given feedback, and if the tools embody the synergy between human educators and AI. Students may exhibit a pre-verbal capacity for philosophical reasoning during learning activities, which may be represented by immature language or illogical thinking. The technical tools here are used to examine the sentences and verify if the logic is complete. It can handle high-demanding and repetitive tasks, allowing

teachers to focus on aspects that necessitate a human-centred perspective, such as extending the unspoken part with students through real-life experiences or social content. Not only teachers but also other participants in class can achieve this as well. This type of tech-assisted, human-guided teaching labor in distribution is likely to be one of the potential patterns of future collaboration between education and AI.

Notes

1. <https://huggingface.co/google-bert/bert-base-chinese>.
2. <https://huggingface.co/ckiplab/bert-base-chinese>.

Disclosure statement

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References

- Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., Avila, R., Babuschkin, I., Balaji, S., Balcom, V., Baltescu, P., Bao, H., Bavarian, M., Belgum, J., ..., & Zoph, B. (2023). Gpt-4 technical report. Preprint, arXiv:2303.08774.
- Allan, E. G., & Driscoll, D. L. (2014). The three-fold benefit of reflective writing: Improving program assessment, student learning, and faculty professional development. *Assessing Writing*, 21, 37–55. <https://doi.org/10.1016/j.asw.2014.03.001>
- Anthropic. (2025). Claude 3.7 Sonnet and Claude Code. <https://www.anthropic.com/news/claude-3-7-sonnet>.
- Beltagy, I., Lo, K., & Cohan, A. (2019). SciBERT: A pretrained language model for scientific text. Preprint arXiv:1903.10676.
- Brondani, M. A. (2010). Students' reflective learning within a community service-learning dental module. *Journal of Dental Education*, 74(6), 628–636. <https://doi.org/10.1002/j.0022-0337.2010.74.6.tb04908.x>
- Brookfield, S. D. (2017). *Becoming a critically reflective teacher*.
- Conejo, R., Barros, B., & Bertoa, M. F. (2018). Automated assessment of complex programming tasks using Siette. *IEEE Transactions on Learning Technologies*, 12(4), 470–484. <https://doi.org/10.1109/TLT.2018.2876249>
- Cui, Y., Che, W., Liu, T., Qin, B., Wang, S., & Hu, G. (2020). Revisiting pre-trained models for Chinese natural language processing. Preprint, arXiv:2004.13922.
- Das, A., Fathallah, N., & Obretincheva, N. (2024). *Navigating nulls, numbers and numerous entities: Robust knowledge base construction from large language models*.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *Bert: Pre-training of deep bidirectional transformers for language understanding*.
- Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., & Mordatch, I. (2023). *Improving factuality and reasoning in language models through multiagent debate*.
- Fu, S., Li, Z., Liu, Z., & Yang, X. (2021). Interactive knowledge distillation for image classification. *Neurocomputing*, 449, 411–421. <https://doi.org/10.1016/j.neucom.2021.04.026>
- Gardner, J., O'Leary, M., & Yuan, L. (2021). Artificial intelligence in educational assessment: 'Breakthrough? Or buncombe and ballyhoo?'. *Journal of Computer Assisted Learning*, 37(5), 1207–1216. <https://doi.org/10.1111/jcal.12577>
- Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., & Yang, A. (2024). The llama 3 herd of models. Preprint, arXiv:2407.21783.
- Insa, D., Prez, S., Silva, J., & Tamarit, S. (2021). Semiautomatic generation and assessment of Java exercises in engineering education. *Computer Applications in Engineering Education*, 29(5), 1034–1050. <https://doi.org/10.1002/cae.22356>
- Katser, M. S., Veremis, B. M., Danciu, T. E., Ramaswamy, V., & Popov, V. (2026). Rallying for reflection: Pilot Use of rubric to facilitate self-reflection in dental education. *European Journal of Dental Education*, 30(1), 31–38. <https://doi.org/10.1111/eje.13088>

- Khlaif, Z., Odeh, A., & Bsharat, T. R. K. (2024). Generative AI-powered adaptive assessment. In: *Power of Persuasive Educational Technologies in Enhancing Learning*, 157–176. IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3693-6397-3.ch007>
- Kong, A., Zhao, S., Chen, H., Li, Q., Qin, Y., Sun, R., et al. (2024). Better zero-shot reasoning with role-play prompting. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 4099–4113).
- Larsen, D. P., London, D. A., & Emke, A. R. (2016). Using reflection to influence practice: Student perceptions of daily reflection in clinical education. *Perspectives on Medical Education*, 5(5), 285–291. <https://doi.org/10.1007/S40037-016-0293-1>
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics (Oxford, England)*, 36(4), 1234–1240. <https://doi.org/10.1093/bioinformatics/btz682>
- Li, T., Li, J., Liu, Z., & Zhang, C. (2020). Few sample knowledge distillation for efficient network compression.
- Liebenberg, J., & Pieterse, V. (2018). Investigating the feasibility of automatic assessment of programming tasks. *Journal of Information Technology Education: Innovations in Practice*, 17, 201–223.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). Roberta: A robustly optimized Bert pretraining approach. Preprint, arXiv:1907.11692.
- Liu, Z., He, X., Liu, L., Liu, T., & Zhai, X. (2023). Context matters: A strategy to pre-train language model for science education.
- Mattfeld, R. S. (2023). Improving student motivation through an alternative grading system. *Journal of Computing Sciences in Colleges*, 39(5), 86–95.
- Messer, M., Brown, N. C. C., Klling, M., & Shi, M. (2024). Automated grading and feedback tools for programming education: A systematic review. *ACM Transactions on Computing Education*, 24(1), 1–43. <https://doi.org/10.1145/3636515>
- Ndukwe, I. G., Daniel, B. K., & Amadi, C. E. (2019). A machine learning grading system using chatbots.
- Ouyang, F., Dinh, T. A., & Xu, W. (2023). A systematic review of AI-driven educational assessment in STEM education. *Journal for STEM Education Research*, 6(3), 408–426. <https://doi.org/10.1007/s41979-023-00112-x>
- Pilicita, A., & Barra, E. (2023). Using of transformers models for text classification to mobile educational applications. *IEEE Latin America Transactions*, 21(6), 730–736. <https://doi.org/10.1109/TLA.2023.10172138>
- Rogers, J., Peecksen, S., Douglas, M., & Simmons, M. (2019). Validation of a reflection rubric for higher education. *Reflective Practice*, 20(6), 761–776. <https://doi.org/10.1080/14623943.2019.1676712>
- Ryan, M. (2013). The pedagogical balancing act: Teaching reflection in higher education. *Teaching in higher education*, 18(2), 144–155. <https://doi.org/10.1080/13562517.2012.694104>
- Su, H., Shi, W., Shen, X., Xiao, Z., Ji, T., Fang, J., & Zhou, J. (2022). RoCBert: Robust Chinese Bert with multimodal contrastive pretraining.
- Tan, S. Y. (2021). Reflective learning? Understanding the student perspective in higher education. *Educational Research*, 63(2), 229–243. <https://doi.org/10.1080/00131881.2021.1917303>
- Team, G., Kamath, A., Ferret, J., Pathak, S., Vieillard, N., Merhej, R., Perrin, S., Matejovicova, T., Ramé, A., Rivière, M., & Rouillard, L. (2025). Gemma 3 technical report. Preprint, arXiv:2503.19786.
- Ullmann, T. D. (2019). Automated analysis of reflection in writing: Validating machine learning approaches. *International Journal of Artificial Intelligence in Education*, 29(2), 217–257. <https://doi.org/10.1007/s40593-019-00174-2>
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824–24837.
- Wu, X., He, X., Liu, T., Liu, N., & Zhai, X. (2023). Matching exemplar as next sentence prediction (MeNSP): Zero-shot prompt learning for automatic scoring in science education. In *Artificial intelligence in education, series = Lecture Notes in Computer Science* (Vol. 13916, pp. 401–413).
- xAI. (2025). Grok 3 Beta –The age of reasoning agents. <https://x.ai/news/grok-3>.
- Xu, Z., & Zhu, P. (2023). Using BERT-based textual analysis to design a smarter classroom mode for computer teaching in higher education institutions. *International Journal of Emerging Technologies in Learning (IJET)*, 18(19), 114–127. <https://doi.org/10.3991/ijet.v18i19.42483>
- Yang, A. C. M., Chen, I. Y. L., Flanagan, B., & Ogata, H. (2021). From human grading to machine grading. *Educational Technology & Society*, 24(1), 164–175.
- Zhang, C., Hofmann, F., Pll, L., & Glser-Zikuda, M. (2024). Classification of reflective writing: A comparative analysis with shallow machine learning and pre-trained language models. *Education and Information Technologies*, 29(16), 21593–21619. <https://doi.org/10.1007/s10639-024-12720-0>
- Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2019). Bertscore: Evaluating text generation with BERT. Preprint, arXiv:1904.09675.
- Zhang, X. (2023). Understanding the role of students' reflections in their uptake of teacher written feedback. *Studies in Educational Evaluation*, 78, 101276. <https://doi.org/10.1016/j.stueduc.2023.101276>